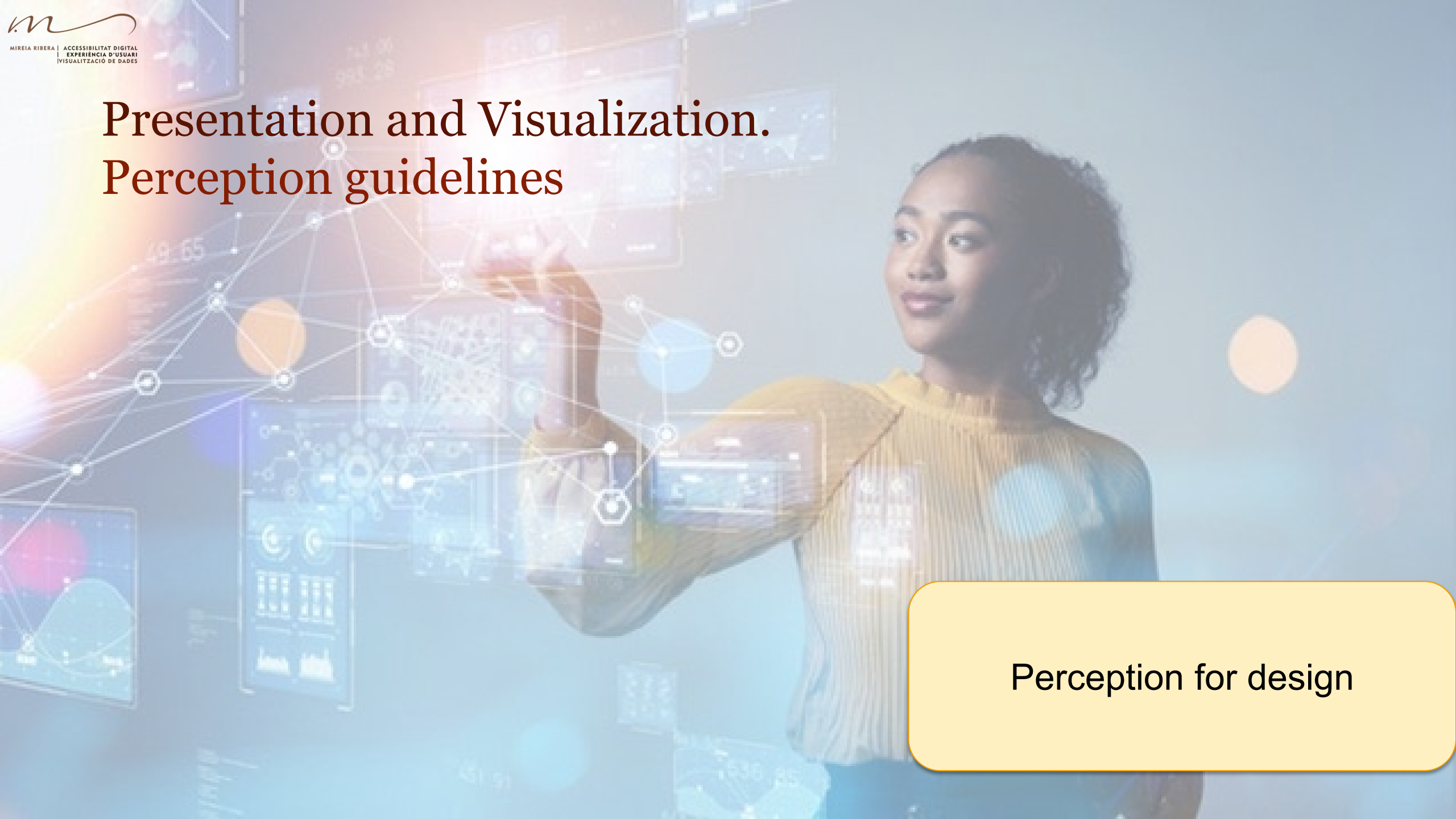
Presentation and Visualization. Perception guidelines

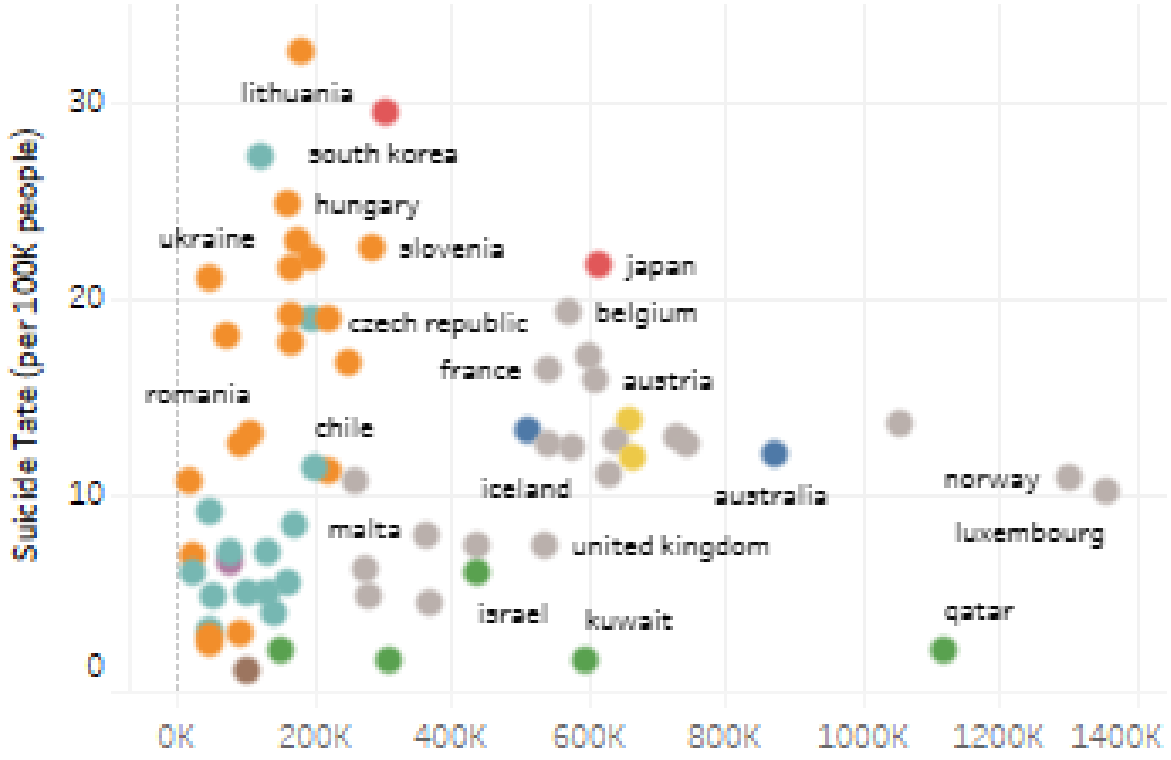


Perception for design

Encoding Guidelines

by Colin Ware “Perception for design”

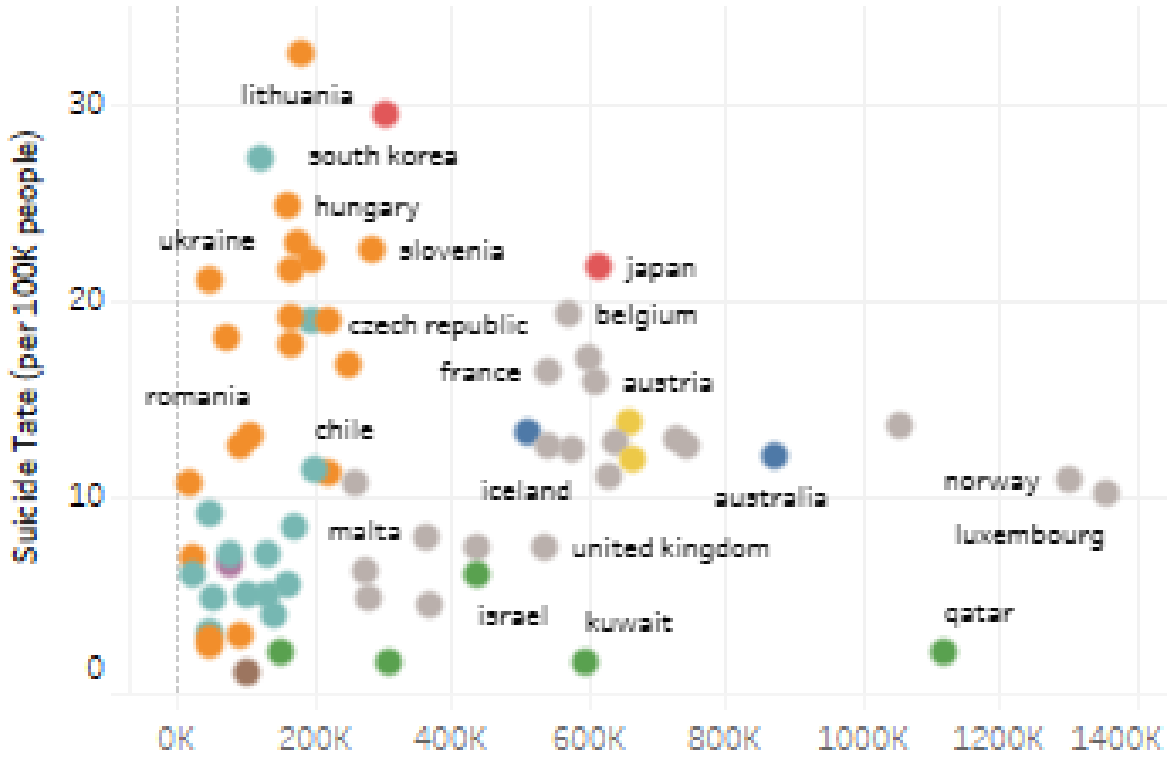
G5.2 “Use different visual channels to display aspects of data so that they are visually distinct”.



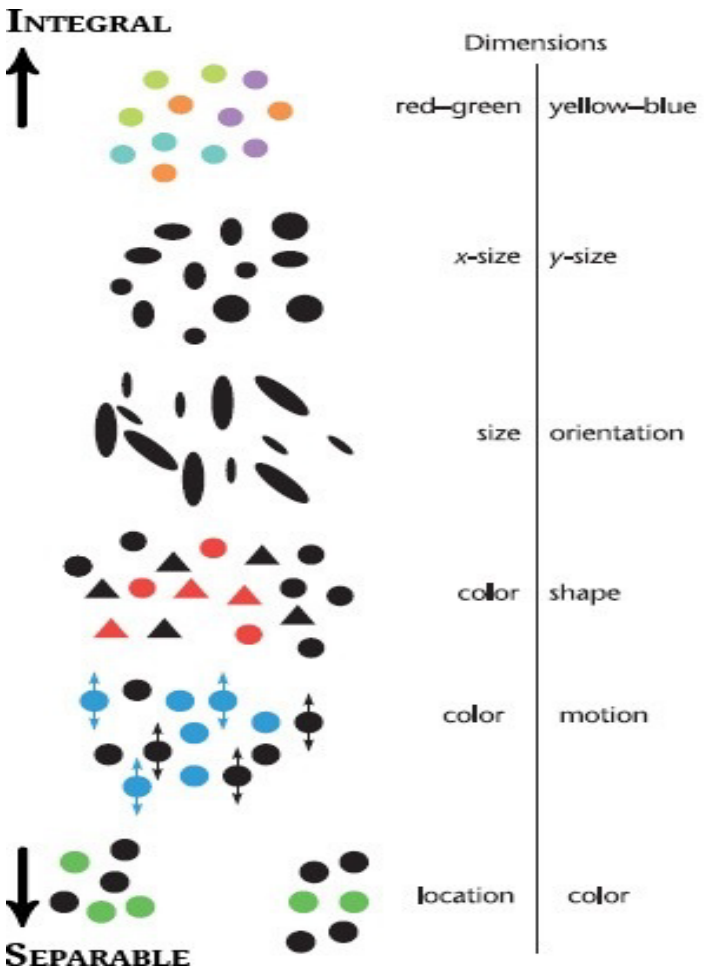
Encoding Guidelines

G5.14 “If it is important for people to respond holistically to a combination of two variables in a set of glyphs, map the variables to integral glyphs properties”.

G5.15 “If it is important for people to respond analytically to a combination of variables, making separate judgments on the basis of one variable or the other, map the variables to separable glyph properties”.



Separable vs Integral Glyph properties



- Integral dimensions are seen together
- Separable dimensions are seen separately

Encoding Guidelines

G5.7 “For maximum pop out a symbol should be the only object in a display that is distinctive on a particular feature channel; for example, it might be the only item that is coloured in a display where everything else is black and white”.

Benefits of slide software:

Visual: Visualizing information helps your readers see what you're explaining.

Versatile: It incorporates photos, illustrations, sketches, and even video if it's posted online.

Interactive: You can embed links and jump around the document itself or out to the Internet.

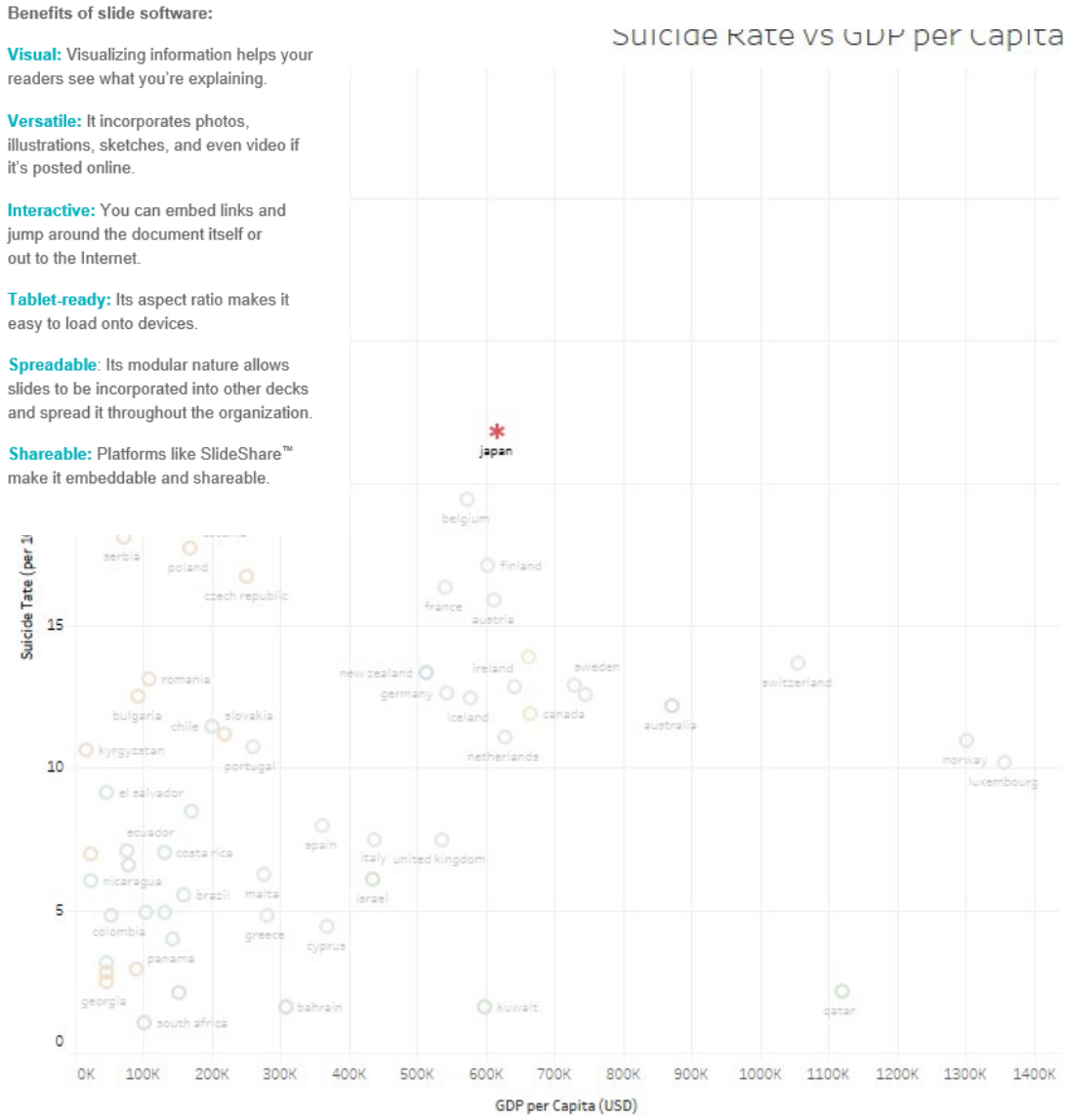
Tablet-ready: Its aspect ratio makes it easy to load onto devices.

Spreadable: Its modular nature allows slides to be incorporated into other decks and spread it throughout the organization.

Shareable: Platforms like SlideShare™ make it embeddable and shareable.

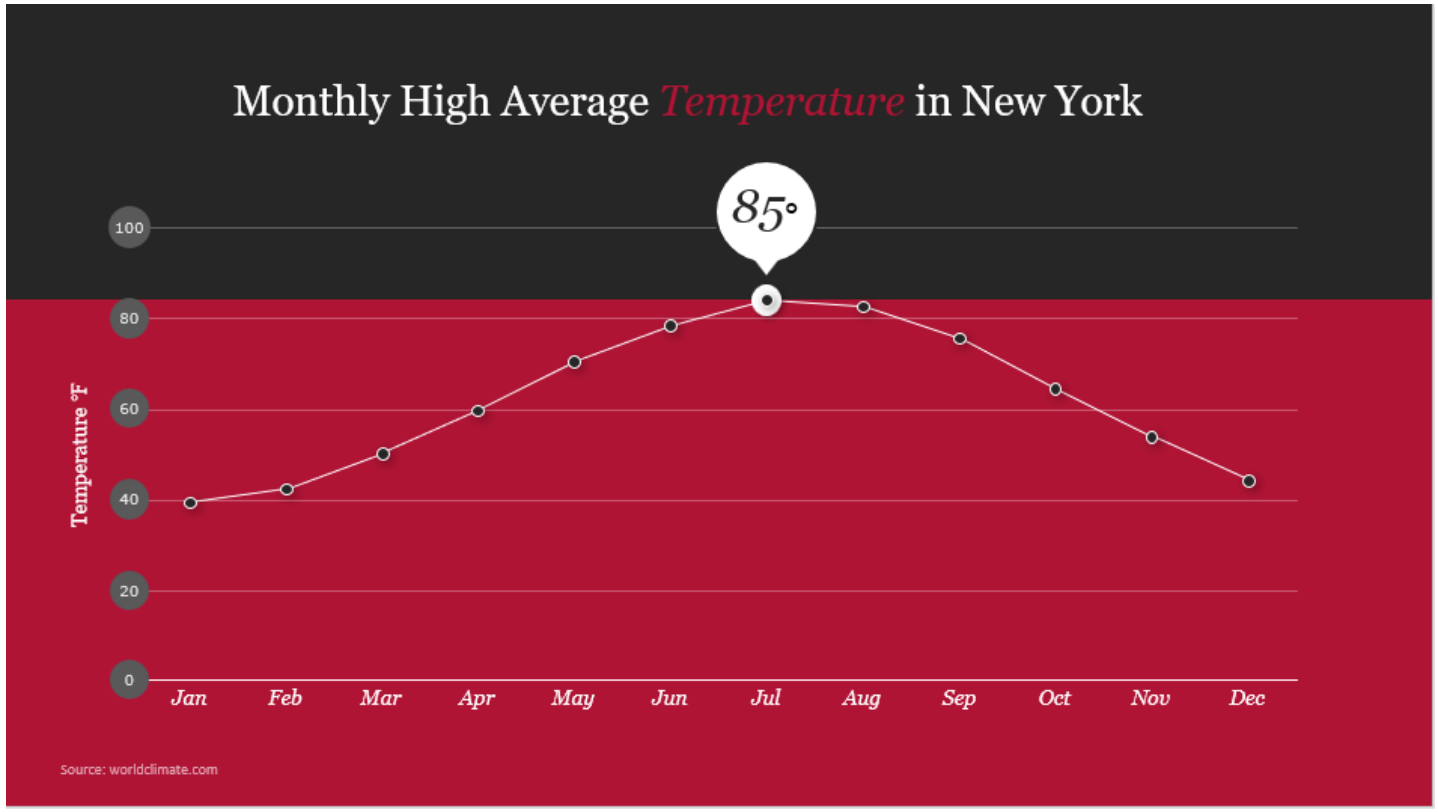
Encoding Guidelines

G5.8 “Use positively asymmetric preattentive cues for highlighting”.



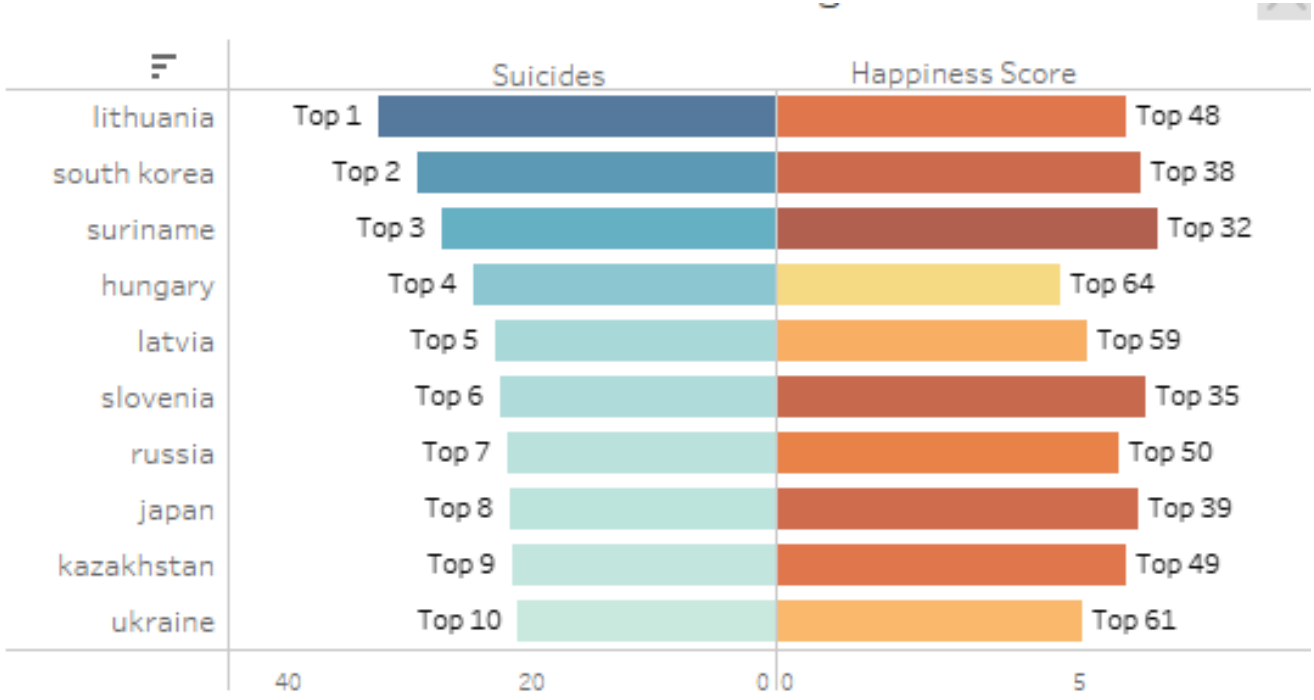
Encoding Guidelines

G5.9 “For highlighting, use whatever feature dimension is used least in other parts of the design”.



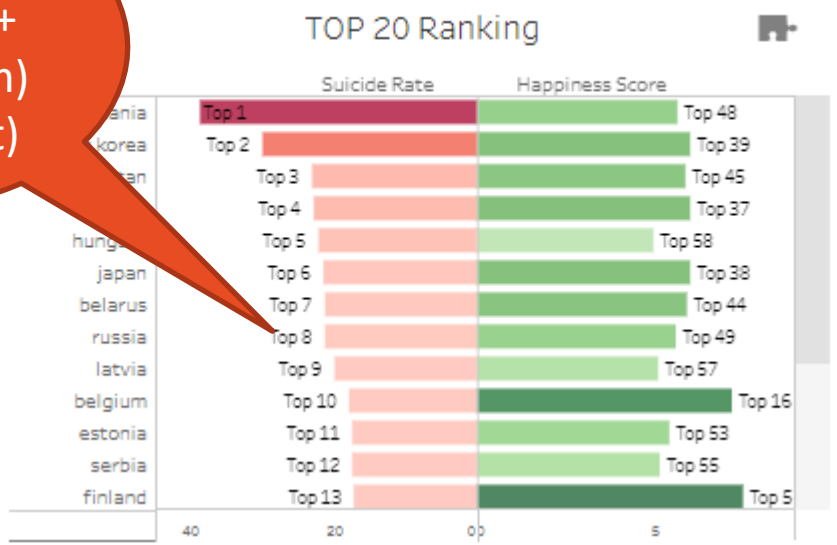
Encoding Guidelines

G5.11 “To make symbols in a set maximally distinctive, use redundant coding wherever possible; for example, make symbols differ in both shape and color”.

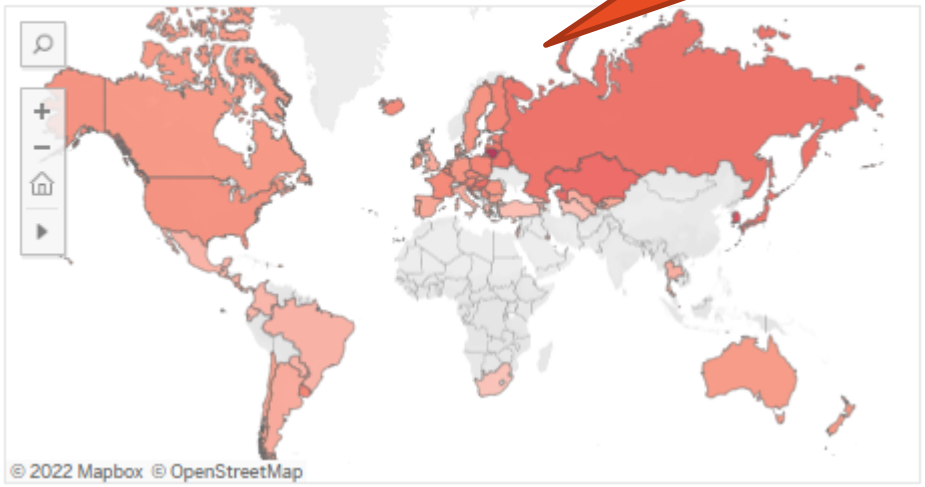


Presentation and Visualization

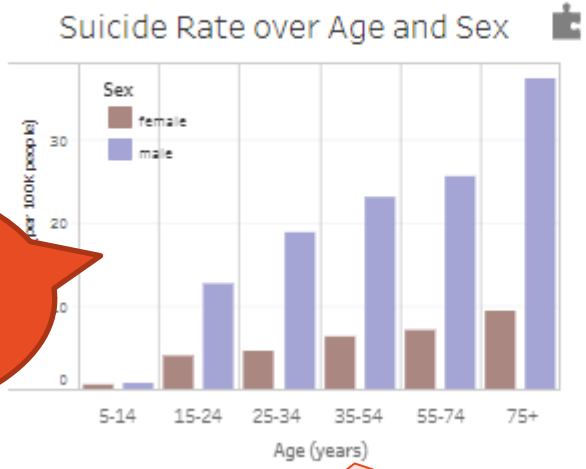
Color (Hue + Intensity) + length(Form) (redundant)

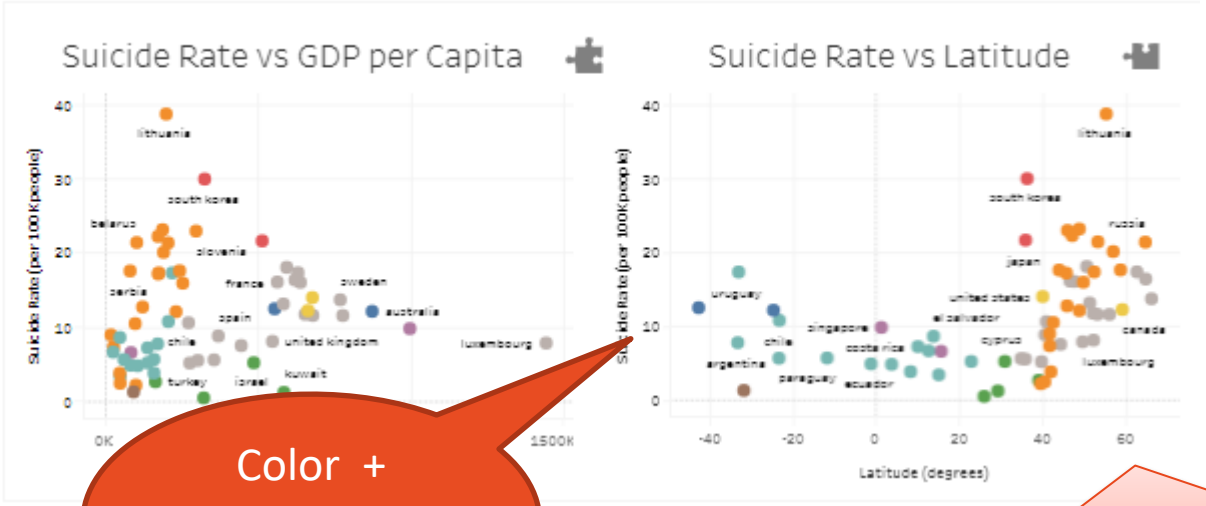


Color (Saturation + Luminosity) = Intensity



Color + length (Form) (separable)



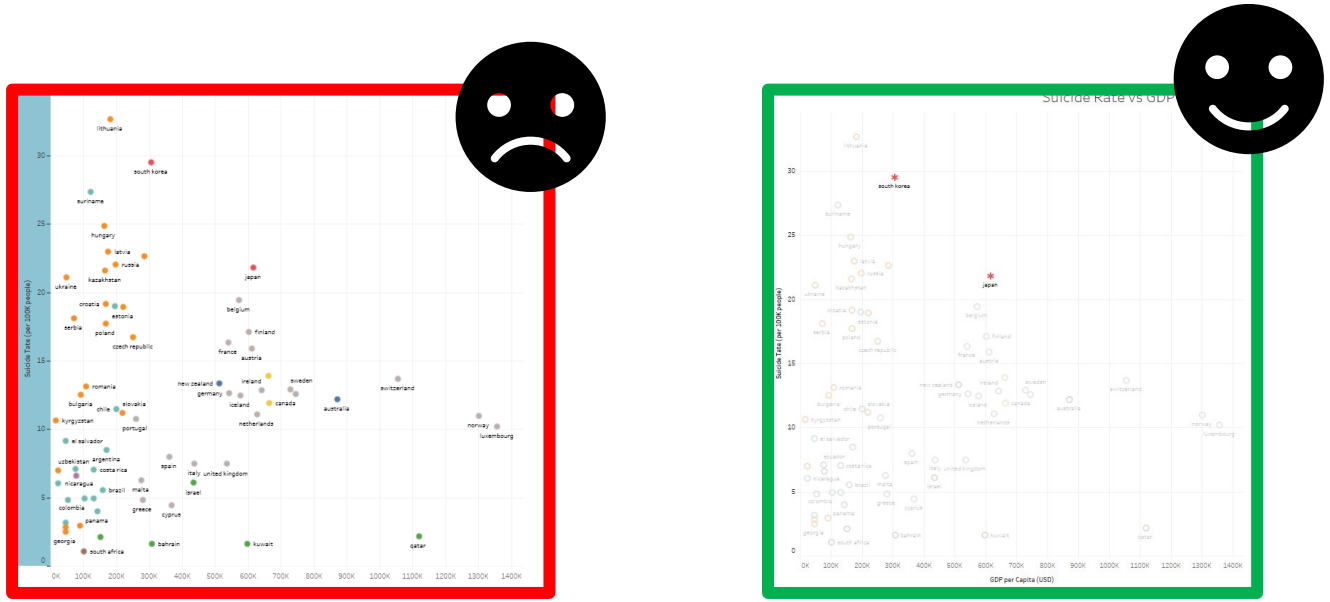


Color + Position (separable)

What do women…? What age …?

How is Central and Eastern Europe GDP and suicide rate?

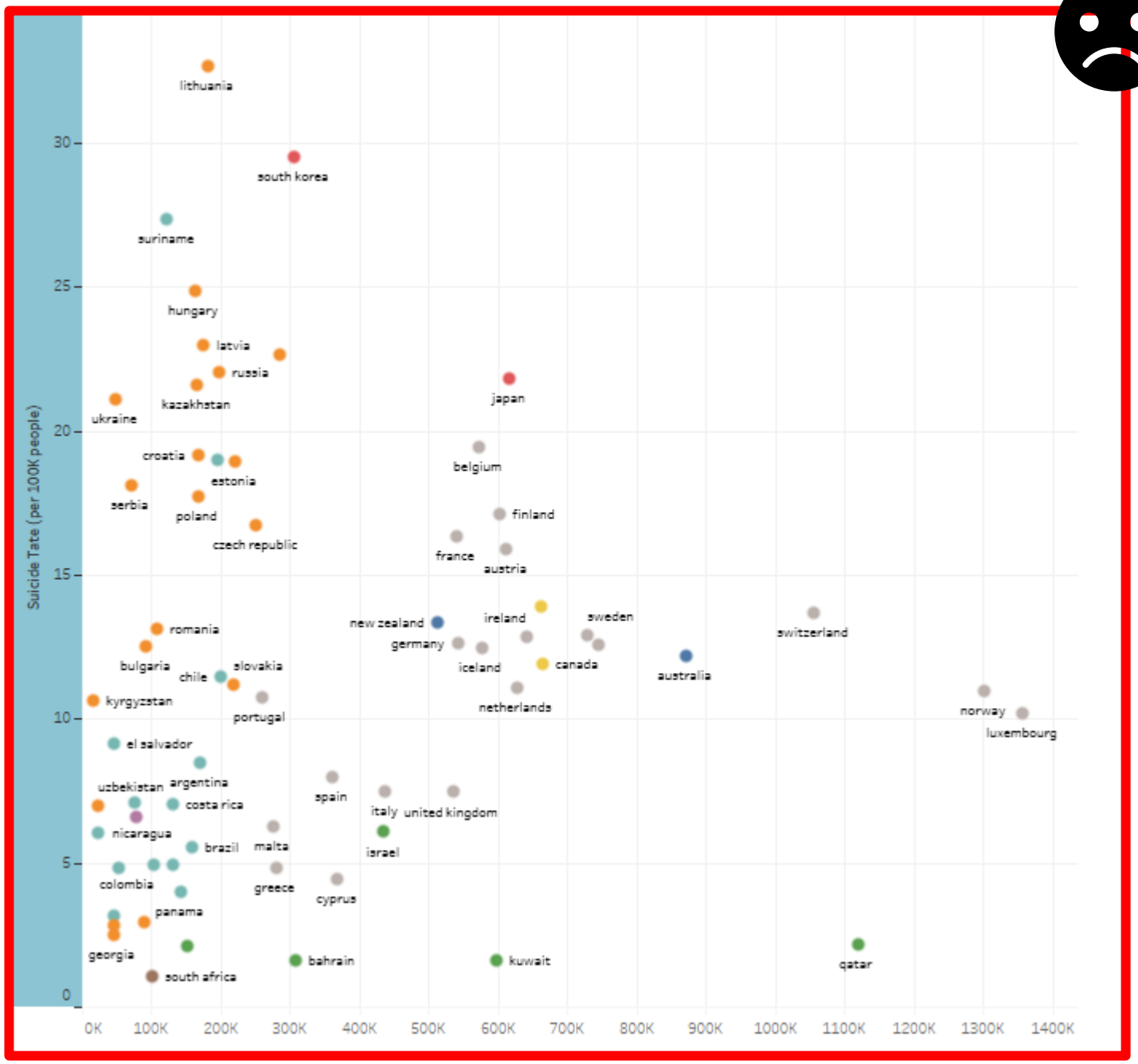
Implementation of guidelines and perception knowledge in charts



Guidelines are exemplified with bad (red) and best (green) practices

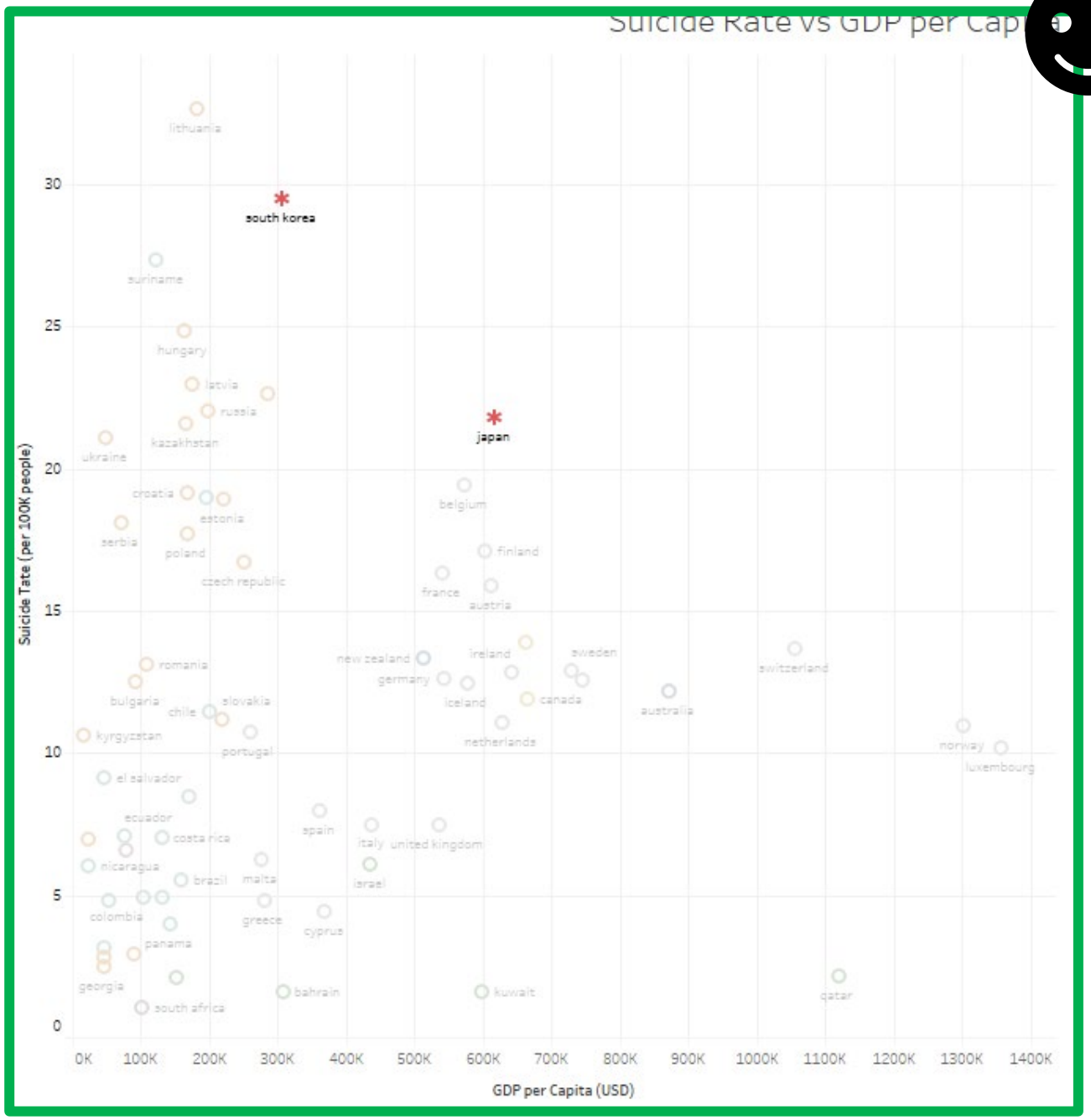
Encoding Guidelines

G5.3 “To make symbols easy to find, make them distinct from their background and from other symbols”.



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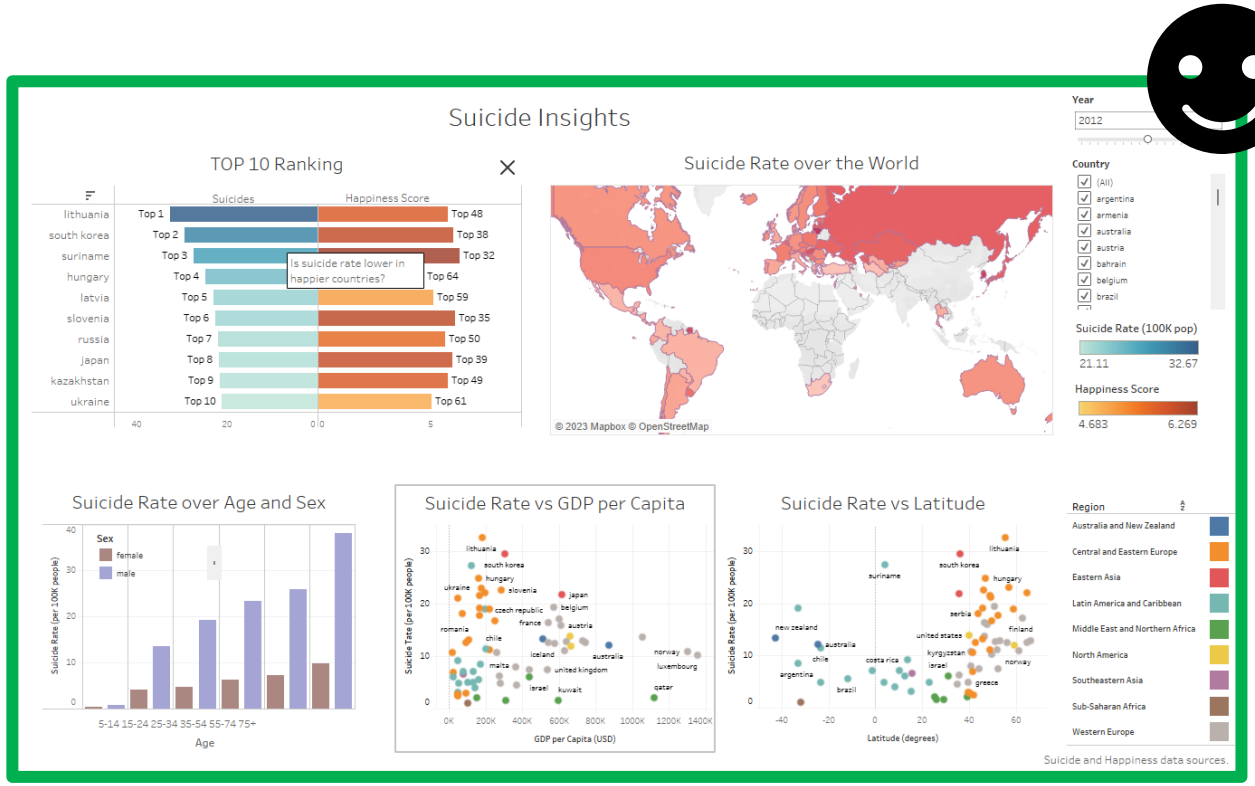
Encoding Guidelines

G6.1 “Place symbols and glyphs representing related information close together”.



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Encoding Guidelines

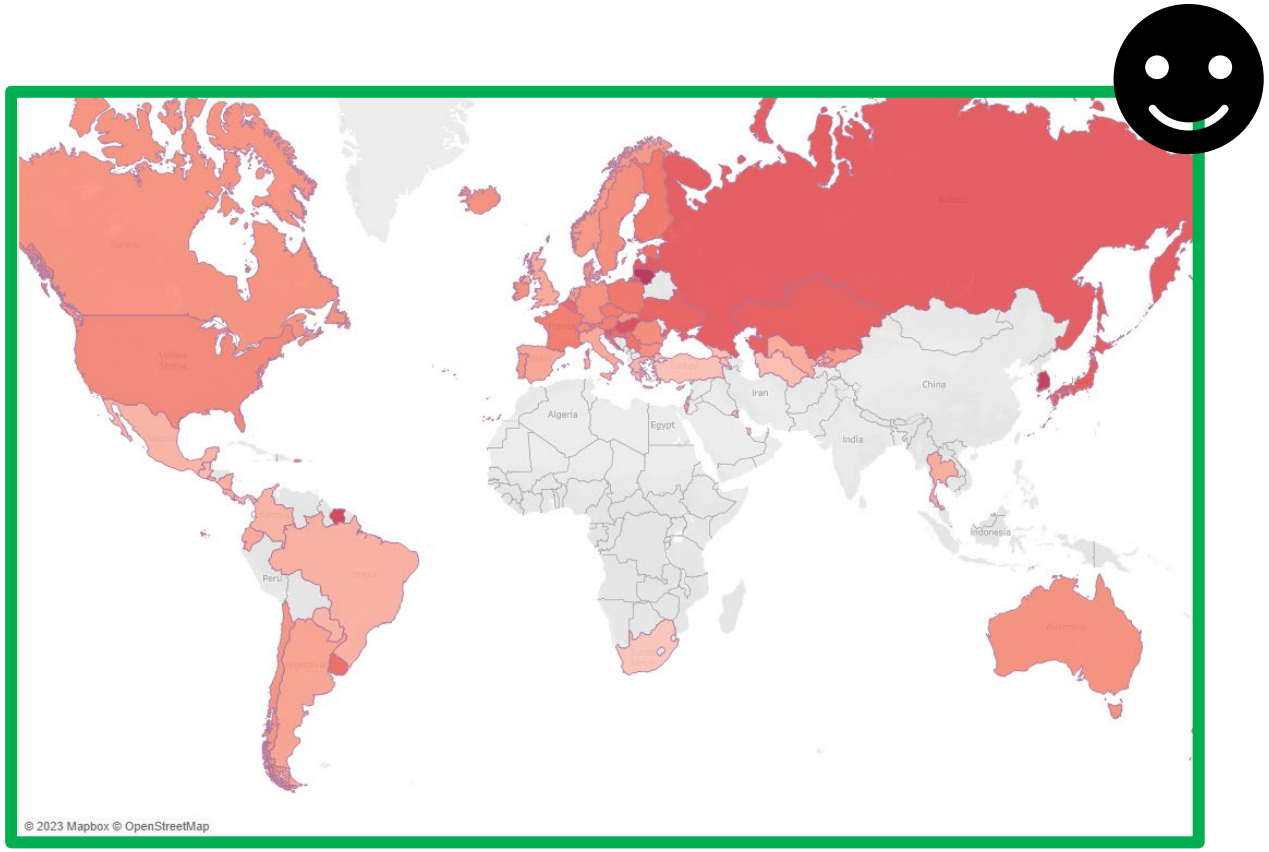


G6.2 “When designing a grid layout of a data set, consider coding rows and/or columns using low-level visual channel properties, such as color and texture”.



Encoding Guidelines

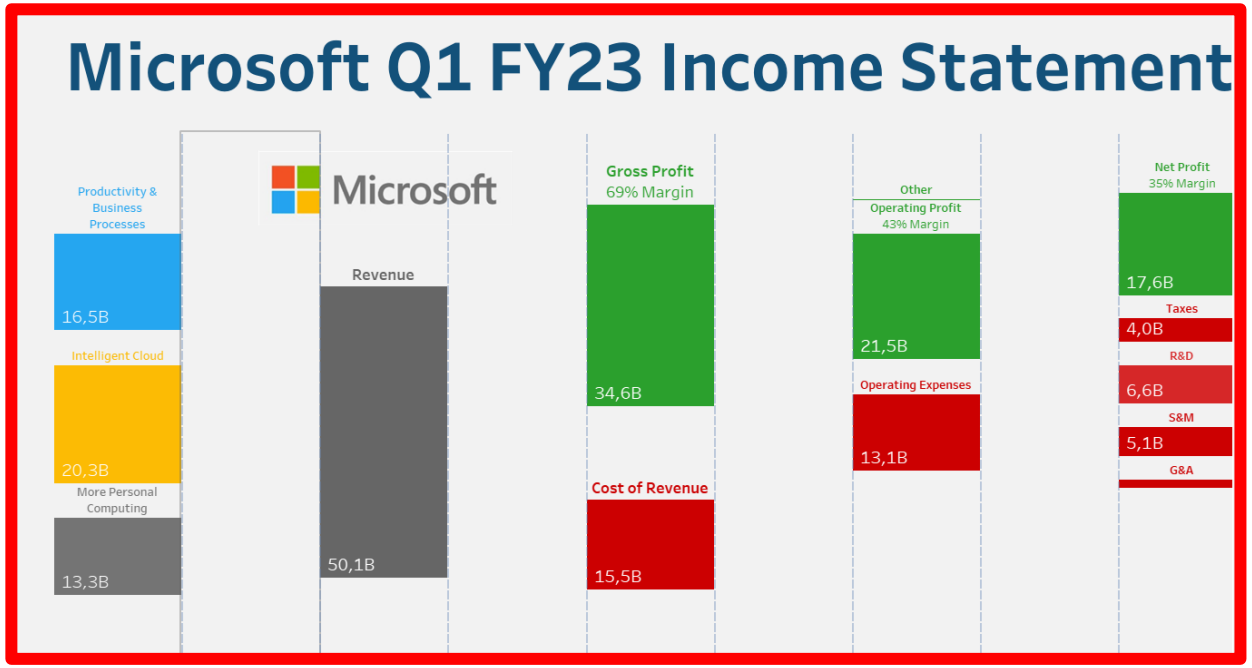
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Encoding Guidelines

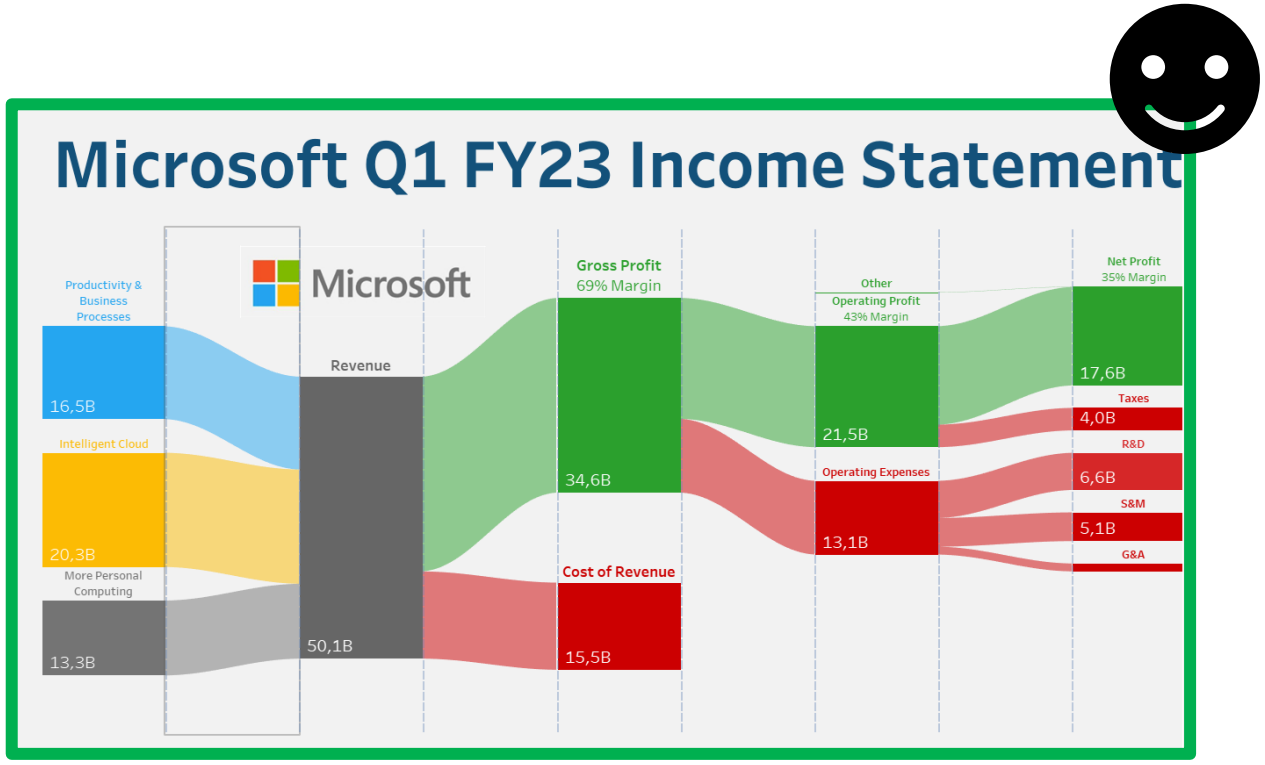


G6.3 “To show relationships between entities, consider linking graphical representations of data objects using lines or ribbons of color”.



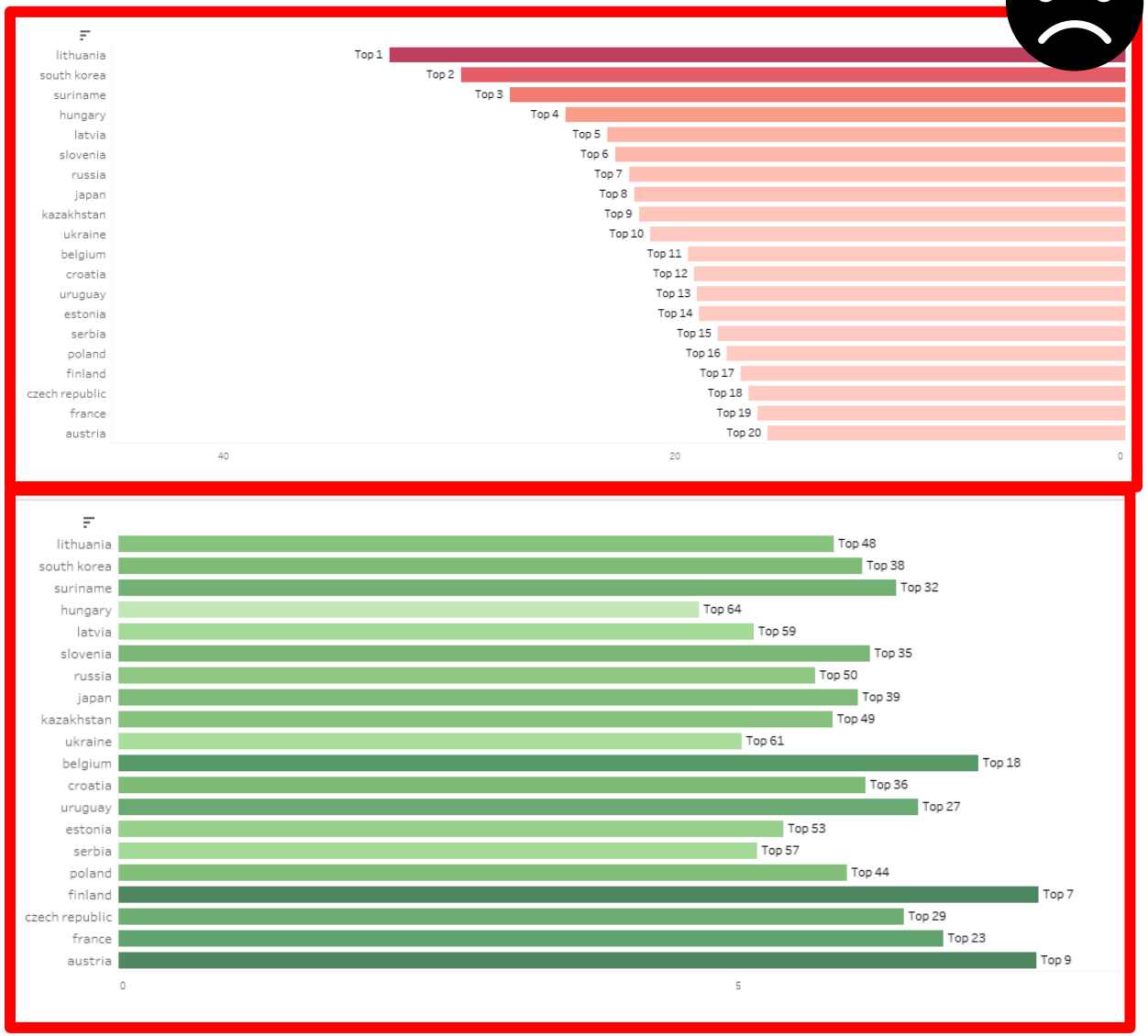
Encoding Guidelines

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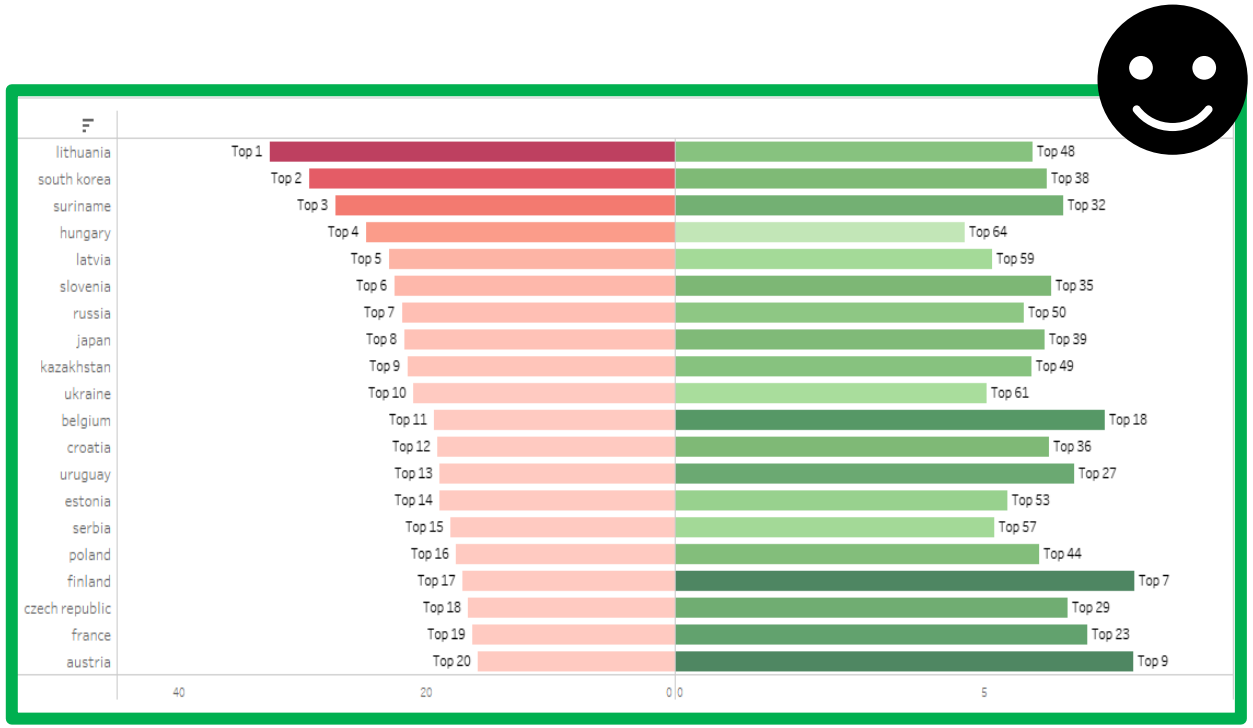
Encoding Guidelines

G6.4 “Consider using symmetry to make pattern comparisons easier. Symmetrical relations should be arranged on horizontal or vertical axes unless some framing pattern is used”.



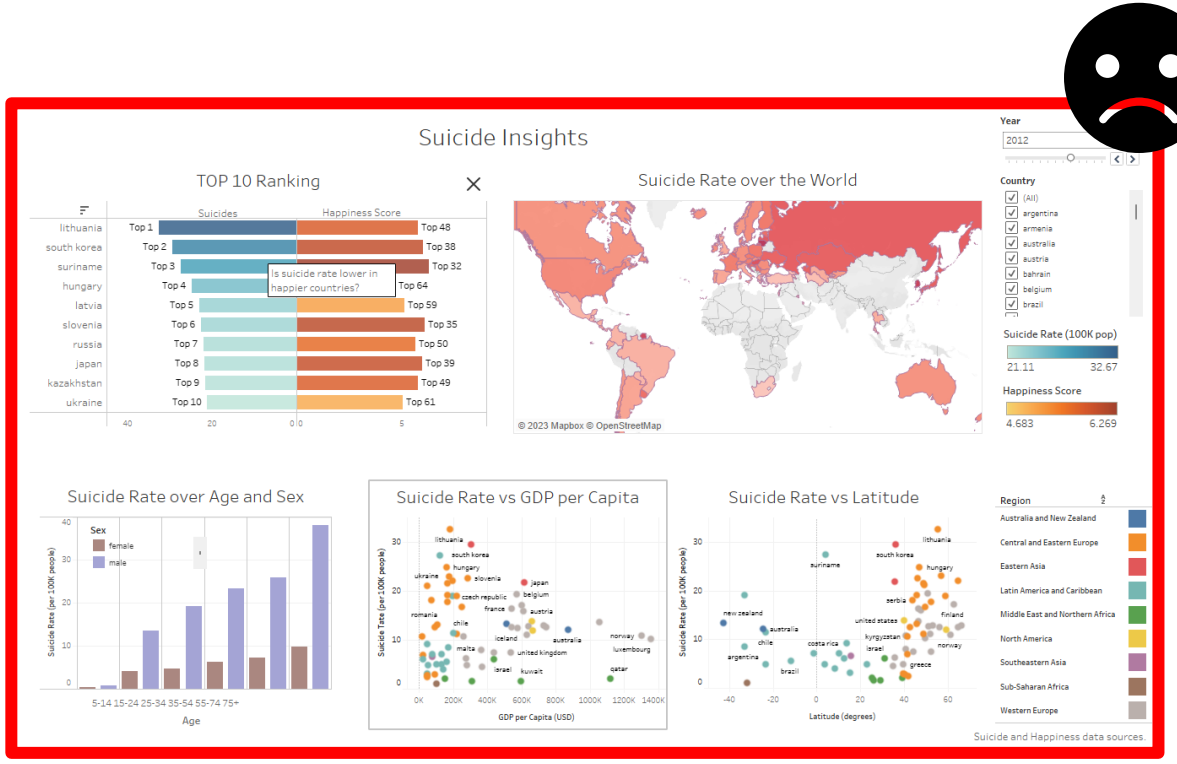
Encoding Guidelines

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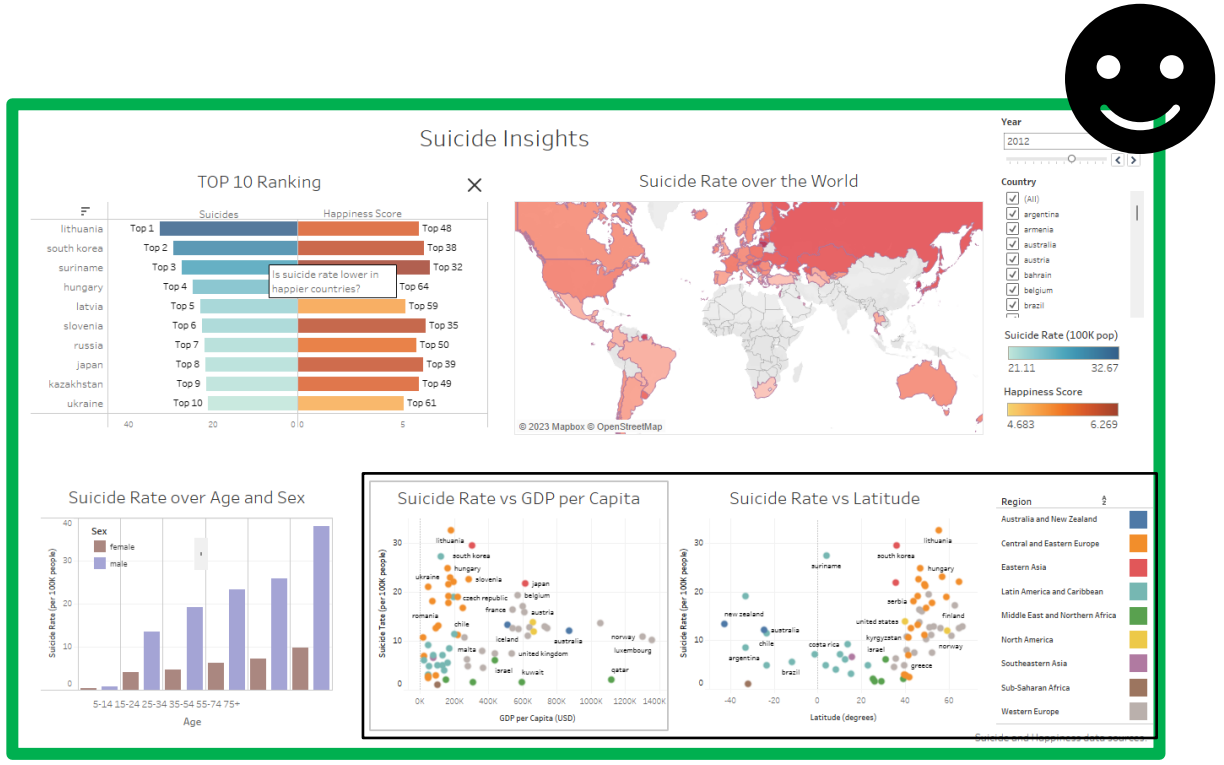
Encoding Guidelines

G6.5 “Consider putting related information inside a closed contour. A line is adequate for regions having a simple shape. Colour or texture can be used to define regions that have more complex shapes”.



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Presentation and Visualization

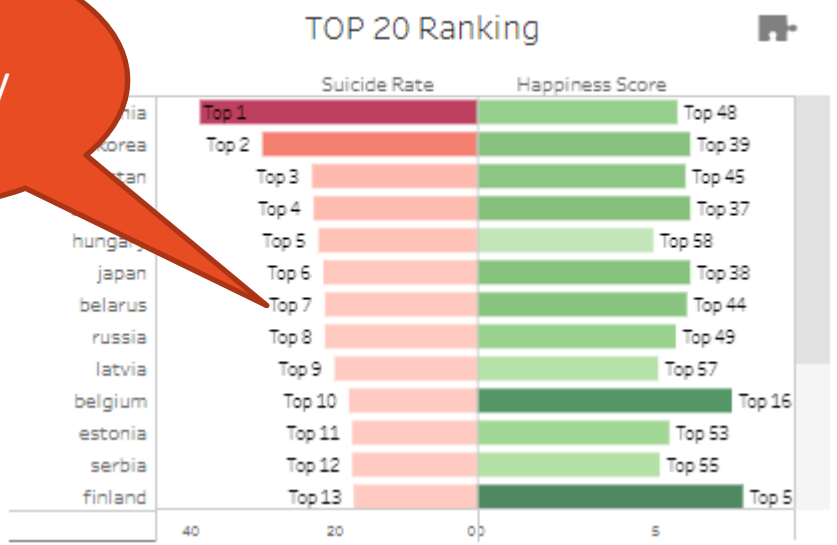
Symmetry

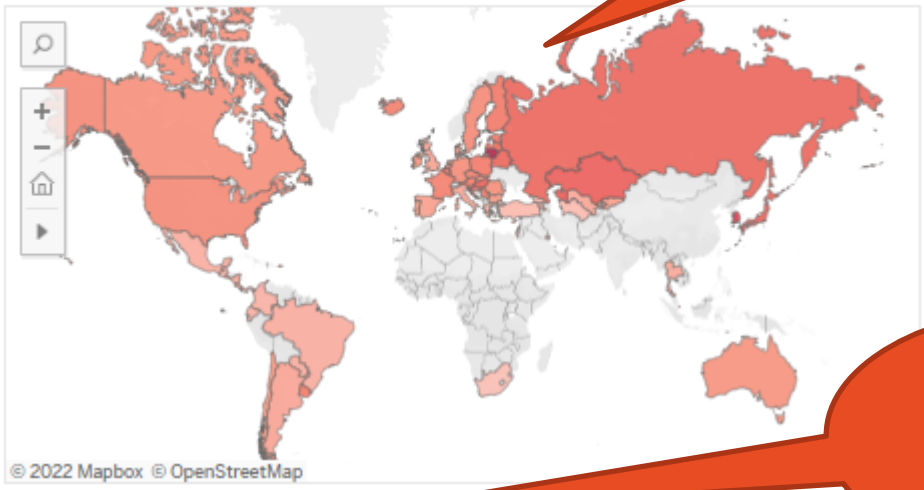
Similarity

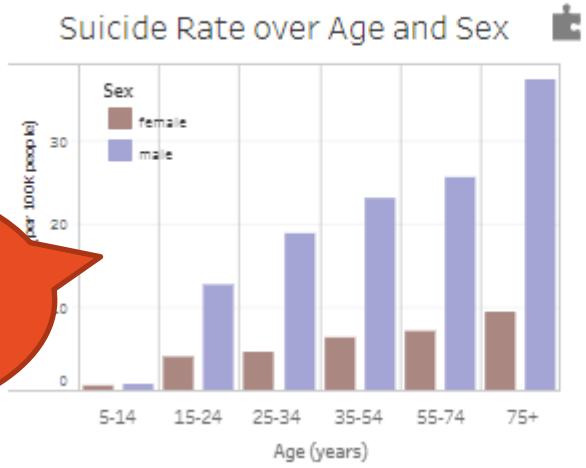
Similarity

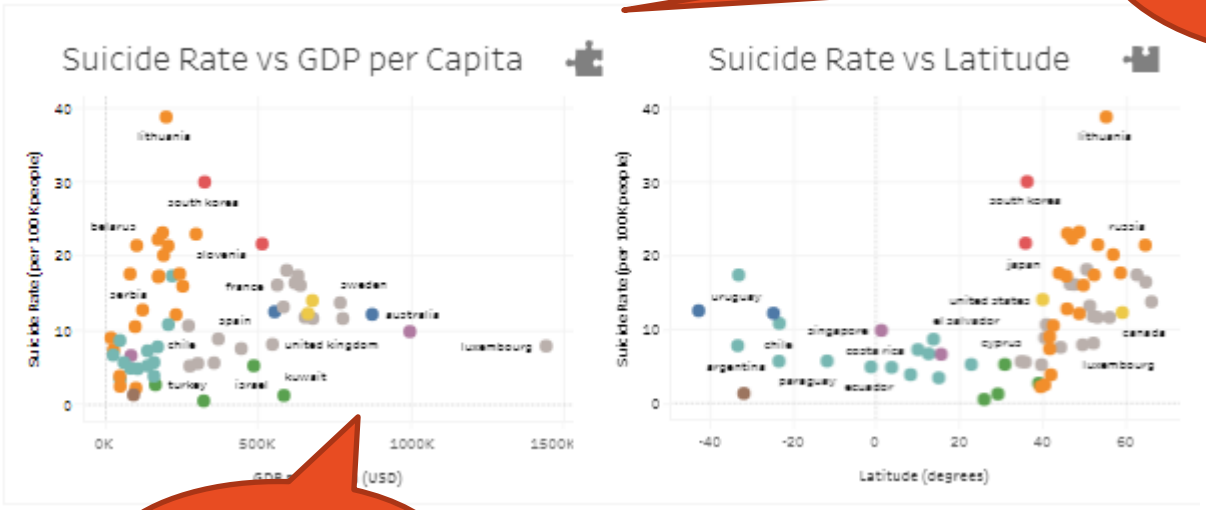
Closure

Proximity









Design principles

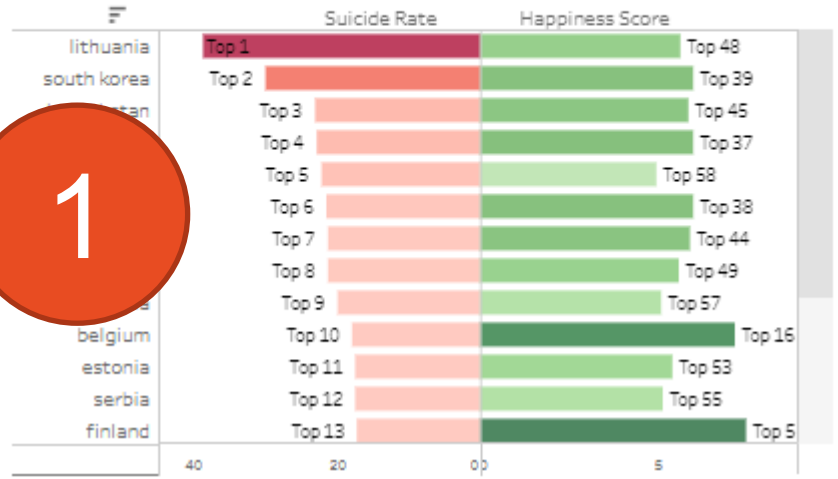
- Important information should receive focused attention, it shall appear in preeminent locations and have visual salience
- Secondary information may be on secondary locations or hidden, only visible on demand

Selective attention and reading patterns



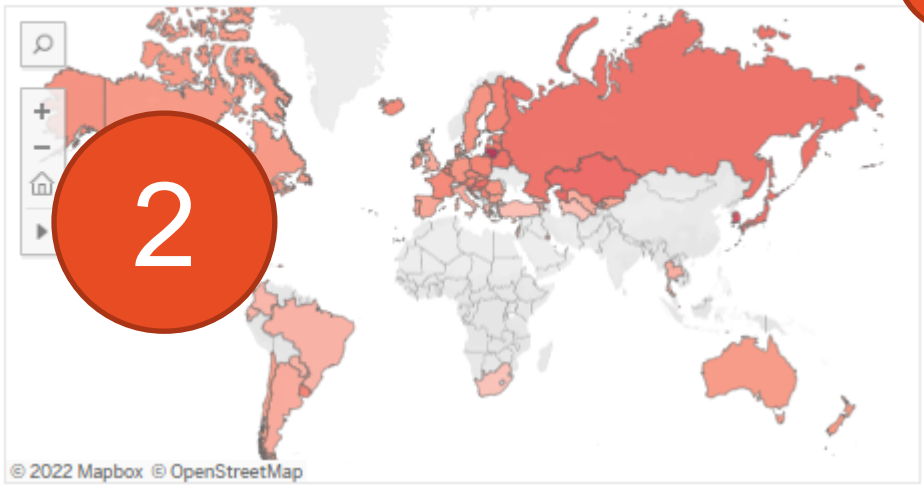
Source: https://www.nngroup.com/articles/banner-blindness-old-and-new-findings/

TOP 20 Ranking



1

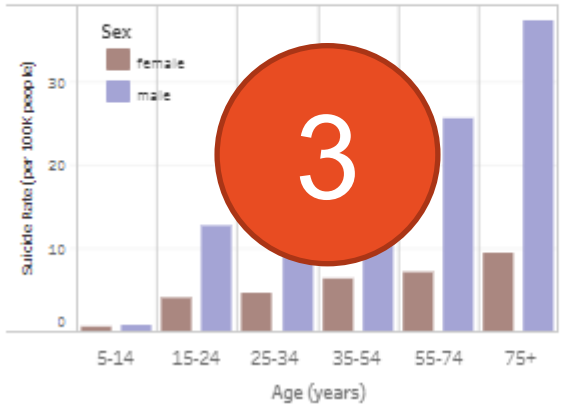
Suicide Rate over the World



2

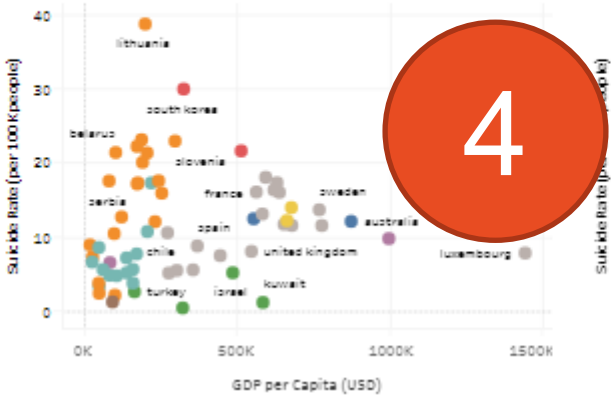
5

Suicide Rate over Age and Sex



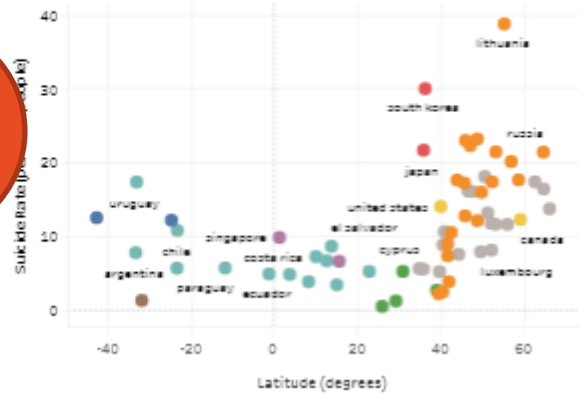
3

Suicide Rate vs GDP per Capita



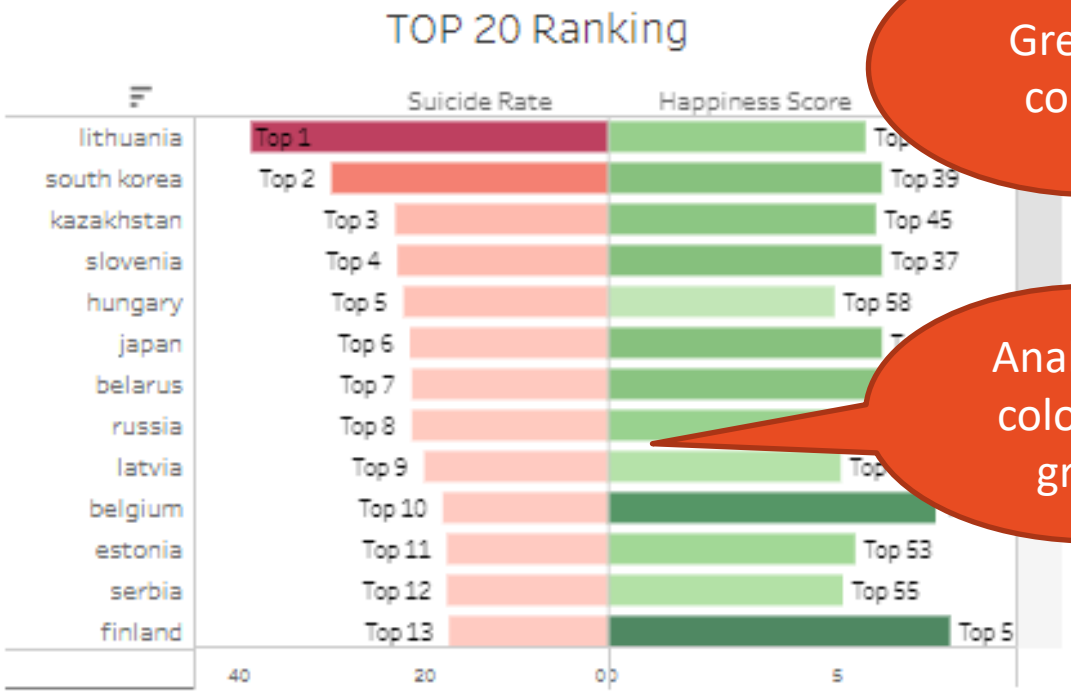
4

Suicide Rate vs Latitude



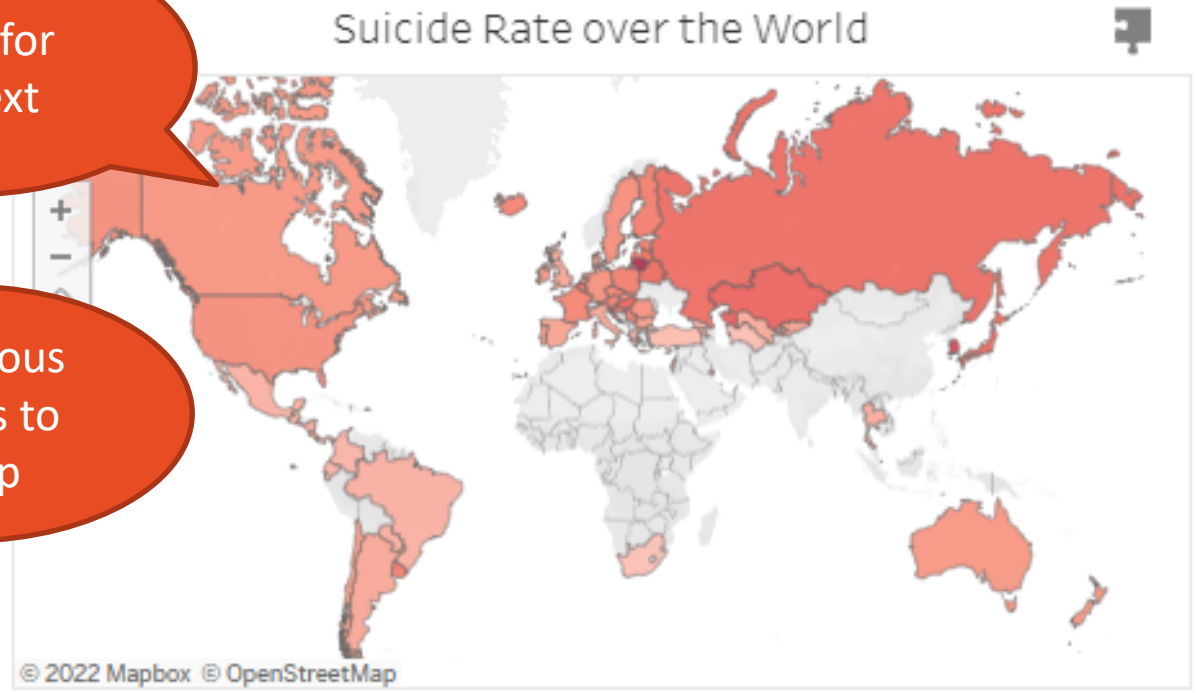
Color for charts

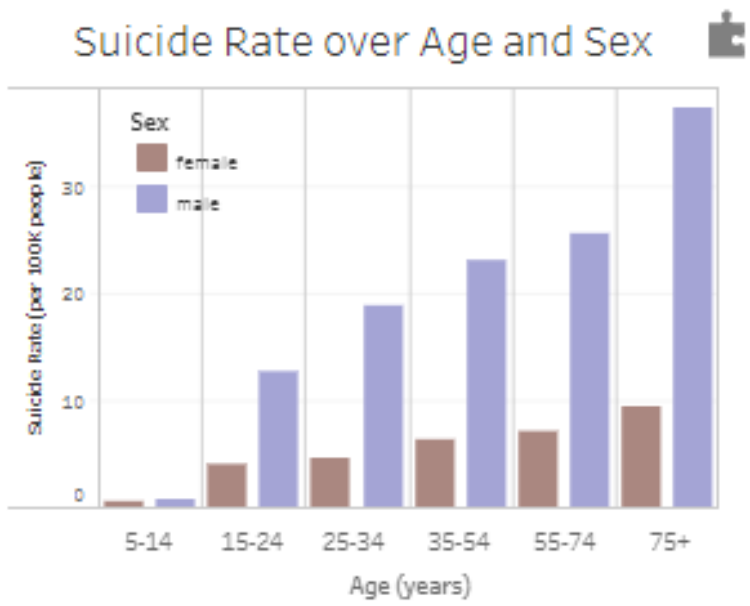
- Assign colour according to function:
  - Use contrast to highlight
  - Analogous colours to group
  - Use greys for context and axis when labelling with colour

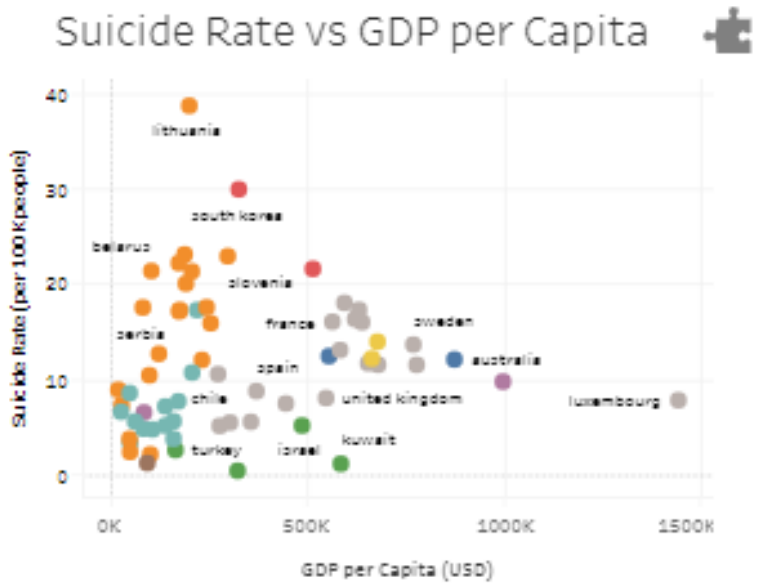


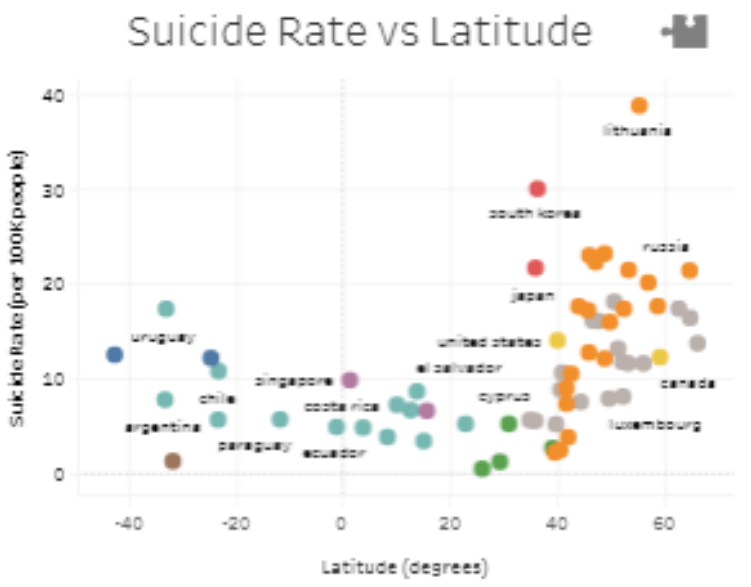
Greys for context

Analogous colours to group

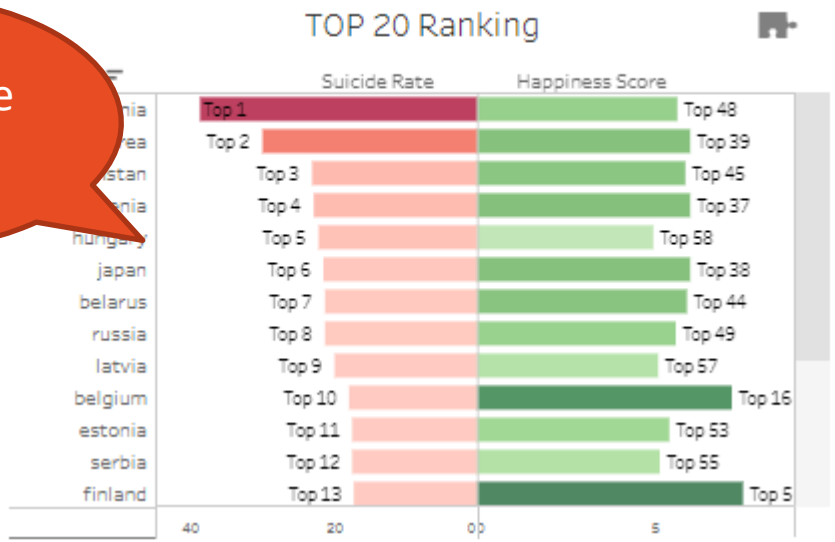




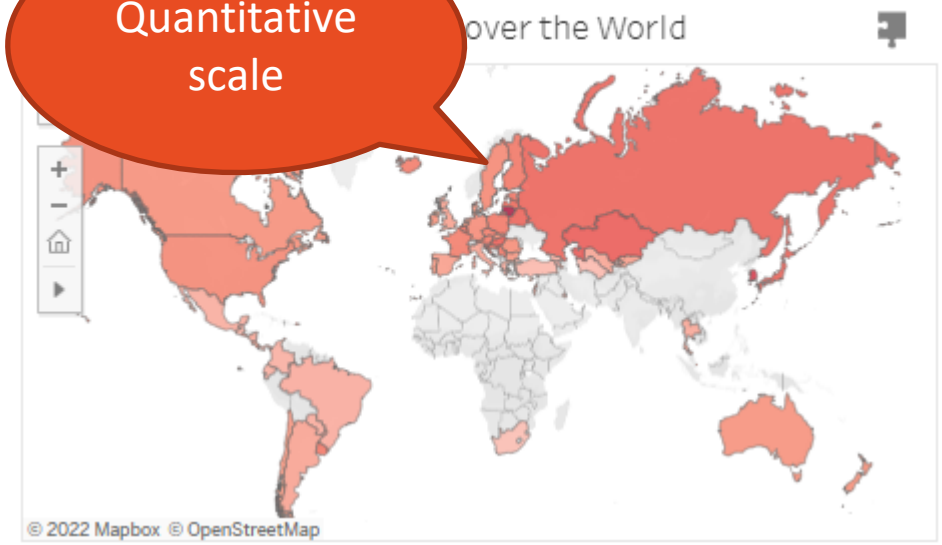


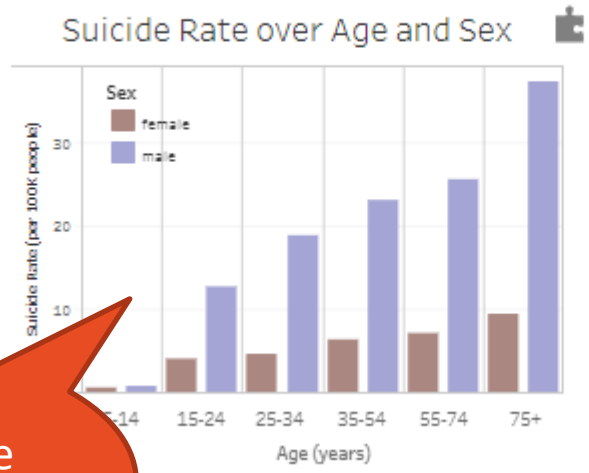


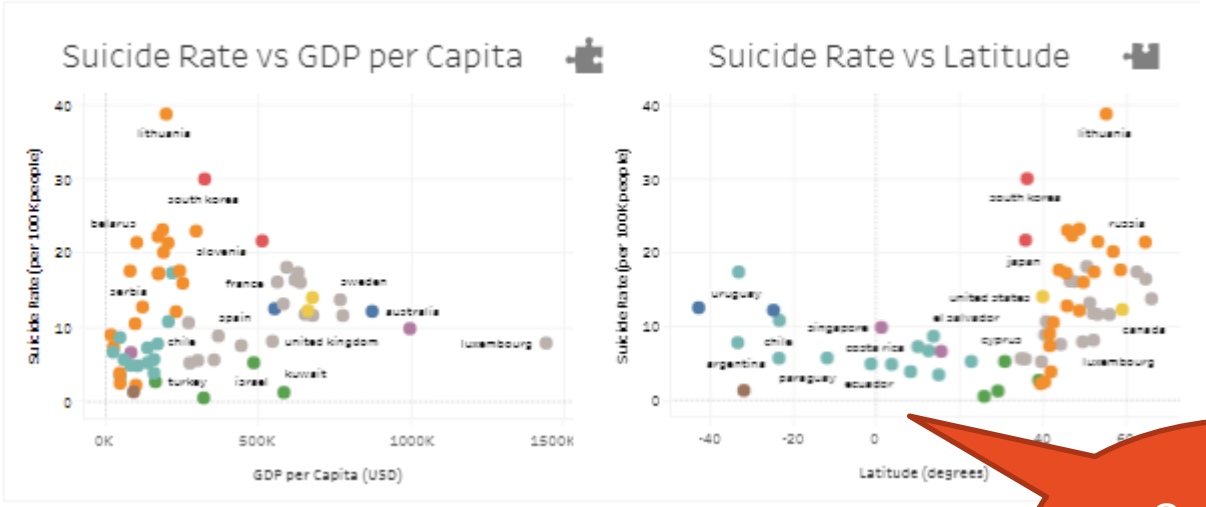
Quantitative scale



Quantitative scale







Qualitative scale

Qualitative scale

Color for maps

- Big areas: low saturation; Small areas: highly saturated
- Ensure hue and luminance contrast with the background (use a border if needed)
- For colour-blindness assure yellow-blue distinction
- See Cynthia Brewer ColorBrewer tool

Key ideas

1. Perception principles should guide our visualization decisions.

2. Encoding, layout, highlighting have their own rules.

3. The correct use of color is very important

Sources

- Ware, C. (2020) Information visualization: perception for design. Burlington: Morgan Kaufmann.
- The example Dashboard on the slides, is made by Muriel Rovira, Jael Freixanet, Emilio Tylson, from course (2021-2022) task.
- Some highlight examples come from Duarte – Slidedocs templates or Animated Charts (www.Duarte.com)

Thank you for your attention

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